Absolute concentration robustness with multistability in bimolecular networks: the exact minimal sizes

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Abstract. Kaihnsa, Nguyen and Shiu proved that an at-most-bimolecular reaction network in which absolute concentration robustness (ACR) and nondegenerate multistationarity coexist has at least 3 species, 3 reactant complexes, 5 complexes and a two-dimensional stoichiometric subspace, all four bounds being attained. They conjectured that each of these four lower bounds becomes strictly larger when multistationarity is replaced by multistability. We determine the four exact minima for ACR with multistability: dimension 3, 3 species, 5 reactant complexes and 6 complexes. The conjecture therefore holds for three of the four quantities and fails for the number of species. Two theorems carry the lower bounds. A two-dimensional bimolecular mass-action system with ACR has at most one exponentially stable positive steady state in each stoichiometric compatibility class; the proof combines a normal form of the vector field on the ACR hyperplane, a connectedness property of the positive steady-state set that holds precisely because products are at most bimolecular, and the global ACR condition. A rank-three system with four reactant complexes and ACR has at most two nondegenerate positive steady states per class, of opposite determinant signs; an exact finite classification reduces this to nineteen scalar normal forms. Six complexes follow by a kinetic-subspace argument and the deficiency-one theorem. Two explicit networks attain the bounds. The six-reaction network $\{Z \to Y + Z,\ 2Z \to Z,\ Y \to Y + Z,\ Y + Z \to 2X,\ X + Z \to 2Z,\ 2X \to Y\}$ has ACR in X for all rate constants and, for suitable ones, two exponentially stable positive steady states, with 3 species, 6 complexes and dimension 3. A four-species, six-reaction network with integer rates has 5 reactant complexes. Stability and ACR are certified in exact arithmetic. We do not know whether one network attains all four minima.

The question

Kaihnsa, Nguyen and Shiu [1] study the minimal size of reaction networks in which ACR and multistationarity coexist. Their main result (Theorem 1.2 there) states that an at-most-bimolecular network G admitting a vector of rate constants for which the mass-action system has ACR and is nondegenerately multistationary has at least 3 species, at least 3 reactant complexes, at least 5 complexes and dimension at least 2; the network $\{X_1 + X_2 \to 2X_3,\ X_3 \to X_1,\ 2X_3 \to 2X_2\}$ attains all four bounds. Section 6 of the paper ends with the following conjecture:

> *"In the present work, our interest in multistationarity comes from the fact that it is a necessary condition for multistability. Another interesting direction, therefore, is to investigate conditions for coexistence of ACR and multistability, rather than multistationarity. The "minimal" networks in the current work admit only two positive steady states and are not multistable. Hence, we conjecture that the lower bounds (on dimension and the numbers of species, reactant complexes, and all complexes) for the coexistence of ACR and multistability are strictly larger than the bounds proven here for ACR and multistationarity."*

Theorem 1 (main result)**.** Let G be an at-most-bimolecular reaction network and suppose that for some vector of positive rate constants κ the mass-action system (G, κ) has ACR and is multistable. Then

(1) the stoichiometric subspace of G has dimension at least 3;

(2) G has at least 3 species;

(3) G has at least 5 reactant complexes;

(4) G has at least 6 complexes.

All four bounds are attained: the network $\mathcal{N}_1$ of Theorem 21, which has ACR for every vector of rate constants, attains those on dimension, species and complexes, and the network $\mathcal{N}_2$ of Section 5 attains the one on reactant complexes. Hence the conjecture holds for dimension, reactant complexes and complexes, and fails for the number of species.

	ACR + nondegenerate multistationarity [1]	ACR + multistability exact minimum	attained by
dimension	2	3	$\mathcal{N}_1$
species	3	3	$\mathcal{N}_1$
reactant complexes	3	5	$\mathcal{N}_2$
complexes	5	6	$\mathcal{N}_1$

Table 2 / Minimal sizes of at-most-bimolecular networks. All bounds of the first column are attained by the network $\{X_1 + X_2 \to 2X_3,\ X_3 \to X_1,\ 2X_3 \to 2X_2\}$ of [1].

Part 1 follows from Theorem 15 (Corollary 17), and part 2 from Theorem 4.1 [1], because a multistable system is nondegenerately multistationary. Part 3 combines Theorem 15 with Theorem 24 (Corollary 25). Part 4 is Corollary 19. The lower and upper bounds concern arbitrary numbers of species and reactions, and no claim of simultaneous minimization is made.

This text merges two reports of this repository, each checked by a fresh-context verifier: one on the conjecture (sections on dimension two, complexes and three species), one on the exact number of reactant complexes. Their original versions are kept in the folder `archive/`. The merged text has not yet been re-verified.

1 Setting

A reaction network G on species $X_1, \dots, X_n$ is a finite set of reactions $y \to y'$ between complexes $y, y' \in \mathbb{Z}^n_{\geq 0}$, $y \neq y'$. It is *at most bimolecular* (bimolecular for short) if every complex, reactant or product, satisfies $|y| := y_1 + \cdots + y_n \leq 2$. Given positive rate constants $\kappa = (\kappa_{y \to y'})$, the mass-action system is

$$\dot{x} = f(x) := \sum_{y \to y'} \kappa_{y \to y'}\, x^y\, (y' - y) = \sum_y x^y\, v_y, \qquad v_y := \sum_{y' \,:\, y \to y'} \kappa_{y \to y'} (y' - y), \tag{3}$$

the second sum running over reactant complexes. The stoichiometric subspace is $S = \operatorname{span}\{y' - y\}$, $s = \dim S$ is the dimension of G, and the stoichiometric compatibility

classes are the sets $(x^0 + S) \cap \mathbb{R}^n_{\geq 0}$. A positive steady state x^* is *nondegenerate* if the restriction $Df(x^*)|_S \colon S \to S$ is invertible, and *exponentially stable* if all eigenvalues of $Df(x^*)|_S$ have negative real part; the latter implies the former.

Definition 4. Following Definition 2.6 [1], (G, κ) has *ACR in* X_i if it has a positive steady state and x_i takes the same value $c > 0$ at all positive steady states (in all compatibility classes). We call (G, κ) *multistable* if some compatibility class contains at least two exponentially stable positive steady states. Both properties refer to the same κ.

This is the usual notion of stability in the literature on small multistable networks §2.2 [2]: a steady state is stable if it is nondegenerate and the nonzero eigenvalues of the Jacobian have negative real parts, which is equivalent to our definition. As in Theorem 1.2 [1], the question is existential: a network qualifies if one vector κ gives both properties.

Exponentially stable steady states are nondegenerate, so a multistable system is nondegenerately multistationary and every lower bound of [1] applies to ACR with multistability.

The zero complex is allowed and counts as a complex, also as a reactant complex: Example 4.13 of [1] counts $0 \to X$ among its four reactant complexes. The result on reactant complexes depends on this convention. Arbitrarily many reactions may leave each reactant complex.

Lemma 5 (coefficient cones)**.** Let G be bimolecular and let v_y be the coefficient vector of the monomial x^y in (3). Then $(v_y)_i \geq 0$ for every $i \notin \operatorname{supp}(y)$, and $\sum_i (v_y)_i \leq 0$ whenever $|y| = 2$.

Proof. Each $y' - y$ with $y \to y'$ has $(y' - y)_i = y'_i \geq 0$ for $i \notin \operatorname{supp}(y)$, and $\sum_i (y' - y)_i = |y'| - |y| \leq 0$ if $|y| = 2$, because $|y'| \leq 2$. Both properties pass to nonnegative combinations. □

The first property is the familiar characterization of mass-action vector fields Lemma 2.3 [1]; the second one is where bimolecularity of the *products* enters, and it is essential below (Remark 16).

2 Dimension two

Throughout this section G is bimolecular with $\dim S = 2$, κ is fixed, (G, κ) has ACR in X_1 with value $c > 0$, f is the vector field (3) and $H := \{x \in \mathbb{R}^n : x_1 = c\}$. All positive steady states lie in H. For $x \in H$ we write $x = (c, x')$ with $x' \in \mathbb{R}^{n-1}$.

Lemma 6. If (G, κ) has a nondegenerate positive steady state, then $e_1 \notin S^\perp$.

Proof. Let x^* be a nondegenerate positive steady state and let W be a matrix whose $n - 2$ rows form a basis of $S^\perp$. The map $F(x) = (f(x), Wx) \in S \times \mathbb{R}^{n-2}$ has invertible derivative at x^*: if $Df(x^*)u = 0$ and $Wu = 0$ then $u \in S$, hence $u = 0$. By the inverse function theorem, for every T near Wx^* there is a positive steady state $x(T)$ near x^* with $Wx(T) = T$. If $e_1 \in S^\perp$ we may take e_1^T as a row of W; the corresponding coordinate of T equals $x_1(T)$, which can then be made different from c, contradicting

↳ ACR. □

Under the hypothesis of the lemma, $S \cap e_1^{\perp}$ is one-dimensional; we fix a vector ℓ spanning it. Every compatibility class meets H in a segment parallel to ℓ.

Lemma 7 (the field on the ACR hyperplane)**.** Suppose some compatibility class contains two distinct nondegenerate positive steady states p and q. Then there are $\alpha \in S \backslash \{0\}$ and a polynomial h of degree at most 2 such that $f(x) = h(x)\,\alpha$ for all $x \in H$. Moreover $q = p + \tau\ell$ for some $\tau \neq 0$, and $h_L(t) := h(p + t\ell)$ has degree exactly 2 with simple zeros $t = 0$ and $t = \tau$.

Proof. Choose a basis σ_1, σ_2 of S and write $f = g_1\sigma_1 + g_2\sigma_2$ with polynomials g_1, g_2 of degree at most 2. Since $Df(p)$ maps onto S, the set $f^{-1}(0)$ is, near p, a smooth submanifold of dimension $n-2$, and by ACR its positive points lie in H. Thus the restrictions $g_1|_H$ and $g_2|_H$, viewed as polynomials on $H \cong \mathbb{R}^{n-1}$, vanish on a smooth hypersurface M of H.

We claim that they have a nonconstant common factor (or one of them vanishes identically, in which case the lemma holds with α the other basis vector). By Baire's theorem, an irreducible factor F_1 of $g_1|_H$ vanishes on a nonempty open subset M_1 of M, and an irreducible factor F_2 of $g_2|_H$ vanishes on a nonempty open subset $M_2 \subseteq M_1$. If F_1 and F_2 were not proportional, they would be coprime, their common complex zero set would have dimension at most $n-3$, and so would its real points; but it contains the $(n-2)$-dimensional manifold M_2.

If the common factor F has degree 2, then $g_i|_H = c_i F$ with constants c_i, and $f|_H = F\,(c_1\sigma_1 + c_2\sigma_2)$. If F has degree 1, then $g_i|_H = F\,m_i$ with $\deg m_i \leq 1$. If m_1, m_2 are proportional we conclude as before. Otherwise $\{x \in H : m_1 = m_2 = 0\}$ is empty or an affine subspace of dimension at most $n-3$. A nondegenerate positive steady state y with $F(y) \neq 0$ would have a neighbourhood in $f^{-1}(0)$, of dimension $n-2$, contained in that set; so every nondegenerate positive steady state satisfies $F = 0$. The steady states of the class of p lie on the segment $(p + \mathbb{R}\ell) \cap \mathbb{R}^n_{>0}$, on which F is affine; vanishing at $p \neq q$, it vanishes on the whole segment, so p is not isolated in its class, which contradicts nondegeneracy (nondegenerate steady states are isolated in their class by the inverse function theorem).

Finally, $p, q \in H$ lie in the same class, so $q - p \in S \cap e_1^{\perp} = \mathbb{R}\ell$. The polynomial h_L vanishes at 0 and τ; if its degree were at most one it would vanish identically and p would again not be isolated. Hence $\deg h_L = 2$ and both zeros are simple. Likewise $f|_H \not\equiv 0$ (otherwise p would not be isolated), so the vector α obtained above is nonzero. □

From now on we assume that a class contains two distinct nondegenerate positive steady states p, q, and we use α, h, ℓ, τ and h_L from Lemma 7.

Lemma 8. If $\alpha_1 = 0$, then $Df|_S$ has a positive real eigenvalue at p or at q.

Proof. Then $\alpha \in S \cap e_1^{\perp}$, so $\alpha = \gamma\ell$ with $\gamma \neq 0$. For $x_t := p + t\ell \in H$ we have $f(x_t) = \gamma h_L(t)\ell$, hence $Df(x_t)\ell = \gamma h_L'(t)\ell$: ℓ is an eigenvector with eigenvalue $\gamma h_L'(t)$. Since $h_L'(0)$ and $h_L'(\tau)$ are nonzero with opposite signs, this eigenvalue is positive at p or at q. □

Normal form. If $\alpha_1 \neq 0$ we normalize $\alpha_1 = 1$. Since $\{\alpha, \ell\}$ is a basis of S and $\ell_1 = 0$, we can write uniquely

$$f = \psi\,\alpha + \varphi\,\ell, \qquad \psi = f_1. \tag{9}$$

On H we have $f = h\alpha$, so φ vanishes on H and $\psi|_H = h$. As $\deg\varphi \leq 2$, there is an affine function P with $\varphi = (x_1 - c)\,P$. The positive steady states are the positive solutions of $\psi = 0$ and $(x_1 - c)P = 0$.

Lemma 10 (determinant on the ACR hyperplane)**.** If $\alpha_1 = 1$, then $\det(Df(x)|_S) = -P(x)\,(\nabla\psi(x)\cdot\ell)$ for every $x \in H$.

Proof. On H, $Df = \alpha\,\nabla\psi^{\mathsf{T}} + P\,\ell\,e_1^{\mathsf{T}}$. Hence $Df\,\alpha = (\nabla\psi\cdot\alpha)\alpha + P\ell$ and $Df\,\ell = (\nabla\psi\cdot\ell)\alpha$. In the basis (α, ℓ) the matrix of $Df|_S$ is $\left(\begin{smallmatrix} \nabla\psi\cdot\alpha & \nabla\psi\cdot\ell \\ P & 0 \end{smallmatrix}\right)$, with determinant $-P\,(\nabla\psi\cdot\ell)$. □

Corollary 11. If $\alpha_1 = 1$ and p, q are both exponentially stable, then $P(p)\,P(q) < 0$.

Proof. A 2×2 matrix whose eigenvalues have negative real parts has positive determinant. Here $\nabla\psi(p)\cdot\ell = h_L'(0)$ and $\nabla\psi(q)\cdot\ell = h_L'(\tau)$ are nonzero with opposite signs, so Lemma 10 forces $P(p)$ and $P(q)$ to have opposite signs. □

The remaining steps show that P cannot change sign along the positive steady states. Write $\psi_H(x') := \psi(c, x')$ and

$$V^+ := \{x' \in \mathbb{R}^{n-1}_{>0} : \psi_H(x') = 0\},$$

so that the positive steady states are exactly the points (c, x') with $x' \in V^+$.

Lemma 12 (bimolecularity makes V^+ connected)**.** Assume $\alpha_1 = 1$. Then V^+ is connected, and $\nabla\psi_H \neq 0$ at every point of V^+.

Proof. For $2 \leq j \leq k \leq n$ let a_{jk} be the coefficient of x_jx_k in $\psi = f_1$. Monomials containing x_1 become of degree at most one in x' on H, and $(x_1 - c)P$ contains no monomial x_jx_k with $j, k \geq 2$; hence the quadratic part of ψ_H is $Q(x') = \sum_{j\leq k} a_{jk}x_jx_k$ and the coefficient vector of x_jx_k in f is $a_{jk}\alpha$. By Lemma 5 applied to $y = X_j + X_k$: $a_{jk} \geq 0$ (coordinate 1), and if $a_{jk} > 0$ then $\alpha_i \geq 0$ for all $i \notin \{1, j, k\}$ and $1 + \sum_{i\geq2}\alpha_i \leq 0$, so that $\sum_{i\in\{j,k\}}\alpha_i \leq -1 - \sum_{i\notin\{1,j,k\}}\alpha_i \leq -1$ (a single term when $j = k$). Let $N := \{i \geq 2 : \alpha_i < 0\}$. Every monomial x_jx_k with $a_{jk} > 0$ therefore satisfies $N \subseteq \{j, k\}$ and $N \cap \{j, k\} \neq \emptyset$. Since $h_L(t) = \psi_H(p' + t\ell')$ has degree 2, $Q(\ell') \neq 0$, so $Q \neq 0$ and $1 \leq |N| \leq 2$.

Second, for $i \in N$ the component f_i restricted to H equals $\alpha_i\psi_H$; its monomials not containing x_i have nonnegative coefficients (setting $x_1 = c > 0$ preserves this property of f_i), so the monomials of ψ_H not containing x_i have nonpositive coefficients. Two cases remain.

Case $N = \{a, b\}$. Then $Q = \beta x_ax_b$ with $\beta > 0$, every other monomial of ψ_H lacks x_a or x_b, and $\psi_H = \beta x_ax_b - G_0 - G_ax_a - G_bx_b - \sum_{k\notin\{a,b\}} G_kx_k$ with all $G \geq 0$. Writing $F := G_0 + G_ax_a + \sum_{k\notin\{a,b\}} G_kx_k$, the equation $\psi_H = 0$ reads $x_b(\beta x_a - G_b) = F$. On the open orthant either $F > 0$, and then V^+ is the graph $x_b = F/(\beta x_a - G_b)$ over the convex set $\{\beta x_a > G_b\} \cap \mathbb{R}^{n-2}_{>0}$, with $\partial\psi_H/\partial x_b = \beta x_a - G_b > 0$; or $F \equiv 0$, and then $V^+ = \{x_a = G_b/\beta\} \cap \mathbb{R}^{n-1}_{>0}$ is convex, with $\partial\psi_H/\partial x_a = \beta x_b > 0$.

Case $N = \{a\}$. Then $Q = x_a \sum_{k\geq 2} \beta_k x_k$ with all $\beta_k \geq 0$, not all zero, and $\psi_H = \beta_a x_a^2 + x_a(K + R(x'')) - G(x'')$, where x'' collects the coordinates other than x_1, x_a, $R(x'') = \sum_{k\neq a} \beta_k x_k$, $K \in \mathbb{R}$, and G is affine with nonnegative coefficients. On the open orthant either $G > 0$ or $G \equiv 0$. If $G > 0$ and $\beta_a > 0$, the quadratic in x_a has exactly one positive root, which is simple; if $G > 0$ and $\beta_a = 0$, the positive root $x_a = G/(K + R)$ exists exactly on the convex set $\{K + R > 0\}$, where $\partial\psi_H/\partial x_a = K + R > 0$. In both cases V^+ is the graph of a smooth function of x'' over a convex set. If $G \equiv 0$, then $V^+ = \{\beta_a x_a + K + R(x'') = 0\} \cap \mathbb{R}^{n-1}_{>0}$ is convex, and on it $\nabla\psi_H = x_a \nabla(\beta_a x_a + K + R) \neq 0$, because β_a or some β_k with $k \neq a$ is positive. □

Lemma 13 (global ACR). Assume $\alpha_1 = 1$ and $P(p)P(q) < 0$, and write $P_H(x') := P(c, x')$. Then $Z := \{x' \in \mathbb{R}^{n-1}_{>0} : P_H(x') = 0\}$ is nonempty, and there is $\varepsilon \in \{1, -1\}$ with $\varepsilon\,\psi_H \geq 0$ on Z.

Proof. P_H takes both signs on the convex set $\mathbb{R}^{n-1}_{>0}$, so $Z \neq \emptyset$; in particular P depends on some x_k with $k \geq 2$, and the hyperplane $\Pi := \{P = 0\}$ of $\mathbb{R}^n$ is transverse to H. At a positive point of Π we have $f = \psi\alpha$, so by ACR ψ has no zero on the convex sets $\Pi^{\pm} := \Pi \cap \mathbb{R}^n_{>0} \cap \{\pm(x_1 - c) > 0\}$, and hence has a constant sign $\varepsilon_{\pm}$ on each. Both sets accumulate at every point of $\{c\} \times Z$ (move inside Π along $e_1 - (\pi_1/\pi_k)e_k$, where π_j are the coefficients of P and $\pi_k \neq 0$), so $\varepsilon_+\psi_H \geq 0$ and $\varepsilon_-\psi_H \geq 0$ on Z. If $\varepsilon_+ \neq \varepsilon_-$, then ψ_H vanishes on Z, a nonempty open subset of the hyperplane $\{P_H = 0\}$, so P_H divides ψ_H: $\psi_H = P_H\, A$ with A affine. Since $P_H(p'), P_H(q') \neq 0$, A vanishes at p' and q', hence on the line through them, and $h_L \equiv 0$, contradicting Lemma 7. So $\varepsilon := \varepsilon_+ = \varepsilon_-$ works. □

Lemma 14 (no transversal sign change). In the situation of Lemma 13, let $z \in V^+ \cap Z$. Then $\nabla\psi_H(z) = \mu\,\nabla P_H$ for some $\mu \neq 0$, and z has a neighbourhood U in V^+ on which $-\operatorname{sign}(\mu)\,\varepsilon\,P_H \geq 0$.

Proof. Let $E := \{P_H = 0\}$, with direction space T. The function $\varepsilon\psi_H$ is nonnegative on the relatively open set $Z \subseteq E$ and vanishes at z, so $\nabla\psi_H(z) \perp T$, that is $\nabla\psi_H(z) = \mu\nabla P_H$; and $\mu \neq 0$ by Lemma 12. Let Q be the quadratic part of ψ_H and B its polar form. For $v \in T$, $\psi_H(z + v) = Q(v)$, so $\varepsilon Q \geq 0$ on T. Fix ν with $\nabla P_H \cdot \nu = 1$ and write $w = v + s\nu$ with $v \in T$, $s = P_H(z + w)$. If $z + w \in V^+$,

$$0 = \psi_H(z+w) = \mu s + Q(v) + 2sB(v,\nu) + s^2 Q(\nu), \quad \text{i.e.} \quad s\,(\mu + 2B(v,\nu) + sQ(\nu)) = -Q(v).$$

For w small the bracket has the sign of μ, so s is zero or has the sign of $-\mu\varepsilon$. □

Theorem 15 (dimension two). Let G be an at-most-bimolecular network of dimension 2 and let κ be such that (G, κ) has ACR. Then every stoichiometric compatibility class contains at most one exponentially stable positive steady state. In particular (G, κ) is not multistable.

Proof. Suppose $p \neq q$ are exponentially stable positive steady states in one class; they are nondegenerate. By Lemmas 6 and 7 we have the objects α, ℓ, h; by Lemma 8, $\alpha_1 \neq 0$, and we normalize $\alpha_1 = 1$. By Corollary 11, $P(p)P(q) < 0$. Define $\theta : V^+ \to \{1, -1\}$ by $\theta(x') = \operatorname{sign} P_H(x')$ if $P_H(x') \neq 0$ and $\theta(x') = -\operatorname{sign}(\mu(x'))\,\varepsilon$ if $x' \in Z$, where $\mu(x') = \nabla\psi_H(x') \cdot \nabla P_H/|\nabla P_H|^2$ is continuous and nonzero on $V^+ \cap Z$ (Lemma 14).

Then θ is locally constant: near a point with $P_H \neq 0$ by continuity, and near $z \in V^+ \cap Z$ by Lemma 14 applied on a neighbourhood where μ keeps its sign. As V^+ is connected (Lemma 12), θ is constant, so $P(p)$ and $P(q)$ have the same sign, a contradiction. □

Remark 16 (where the bimolecular products matter)**.** The bound $|y'| \leq 2$ on product complexes is used only through the second property in Lemma 5, inside Lemma 12. Without it, ψ_H may contain the terms $x_2^2 - 3x_2 + 2$, and V^+ may consist of two parallel sheets on which P has opposite signs; the argument then breaks down. In the planar model $\dot{u} = t^2 - 1 - u/c$, $\dot{t} = -ut$ (with $u = x_1 - c$), the two stable nodes $t = \pm 1$ of the ACR line $u = 0$ are separated by the invariant line $t = 0$, which ends at a saddle on the face $x_1 = 0$. We have not found a mass-action realization of this mechanism with bimolecular reactant complexes (linear programs with the constraints of Lemma 5 minus the product bound are feasible locally, but the global ACR condition failed on every sample), so we do not know whether Theorem 15 extends to networks with larger product complexes. ⌟

3 Dimension and number of complexes

Corollary 17 (dimension)**.** If a bimolecular network G admits κ with ACR and multistability, then $\dim S \geq 3$.

Proof. One-dimensional networks never have ACR with nondegenerate multistationarity Proposition 3.10 [1], dimension 2 is excluded by Theorem 15, and dimension 0 has singleton classes. □

For the number of complexes we use the classical objects of chemical reaction network theory: m complexes, ℓ linkage classes, t terminal strong linkage classes, deficiency $\delta = m - \ell - s$, and the factorization $f(x) = Y A_\kappa \Psi(x)$, where Y is the $n \times m$ matrix of complexes, $\Psi(x) = (x^y)_y$ and A_κ is the Laplacian matrix of the weighted reaction graph [3]. The kinetic subspace $K_\kappa := Y(\operatorname{Im} A_\kappa) \subseteq S$ contains every value of f and every column of Df.

Lemma 18 (kinetic subspace)**.** If (G, κ) has a nondegenerate positive steady state and G has a nonterminal complex, then $\delta \geq t - \ell + 1$.

Proof. At a nondegenerate positive steady state x^* the image of $Df(x^*)|_S$ is S, so $K_\kappa = S$. The kernel of A_κ has dimension t and consists of vectors supported on terminal complexes, so $\dim \operatorname{Im} A_\kappa = m - t$, and $A_\kappa \Psi(x^*) \neq 0$ because $\Psi(x^*)$ is positive on a nonterminal complex. As $Y A_\kappa \Psi(x^*) = f(x^*) = 0$, the map Y has a nonzero kernel on $\operatorname{Im} A_\kappa$. Hence $s = \dim K_\kappa \leq m - t - 1$, that is $\delta = m - \ell - s \geq t - \ell + 1$. □

Corollary 19 (complexes)**.** If a bimolecular network G admits κ with ACR and multistability, then G has at least 6 complexes.

Proof. By Corollary 17, $s \geq 3$. A network of deficiency zero is not multistationary Lemma 2.4 [1], so $\delta \geq 1$ and $m = \ell + s + \delta \geq 5$, with equality only if $\ell = 1$, $s = 3$, $\delta = 1$. Suppose $m = 5$. If G has a nonterminal complex, Lemma 18 gives $t \leq \ell + \delta - 1 = 1$; otherwise all complexes are terminal and $t = \ell = 1$. So G has one linkage class, of deficiency one, with a single terminal strong linkage class, and the deficiency-one theorem [4] gives at most one positive steady state in each compatibility class. This ⌟

contradicts multistability. □

Remark 20 (on the deficiency-one theorem)**.** The uniqueness part of the deficiency-one theorem holds without weak reversibility, provided each linkage class contains exactly one terminal strong linkage class, $\delta_\theta \le 1$ for each linkage class and $\sum_\theta \delta_\theta = \delta$ [4]. Kaihnsa, Nguyen and Shiu [Lemma 2.5 [1]] state it for weakly reversible networks only, but their proof of the bound of 5 complexes (Theorem 4.9(2) there) applies it to networks that need not be weakly reversible; Lemma 18 supplies the missing hypothesis $t = 1$ there, so their bound stands. The same argument shows that two nondegenerate positive steady states in one class force $m \ge s+3$ for networks of any molecularity: $\delta \ge 1$ always, and $\delta \ge 2$ when $\ell = 1$.

4 Three species and six complexes suffice

Theorem 21 (three species, six complexes)**.** The bimolecular network on species X, Y, Z

$$\mathcal{N}_1: \quad Z \xrightarrow{k_1} Y+Z, \quad 2Z \xrightarrow{k_2} Z, \quad Y \xrightarrow{k_3} Y+Z,$$
$$Y+Z \xrightarrow{k_4} 2X, \quad X+Z \xrightarrow{k_5} 2Z, \quad 2X \xrightarrow{k_6} Y$$

has 3 species, 6 reactions, 6 reactant complexes, 6 complexes and dimension 3. For every vector of rate constants it has ACR in X, with value $2k_1/k_5$. For $(k_1,\dots,k_6) = (5,1,1,4,10,7)$ its positive steady states are exactly

$$P_1 = (1, \tfrac{19}{4}, \tfrac{1}{2}), \qquad P_2 = (1,3,1), \qquad P_3 = (1, \tfrac{7}{4}, \tfrac{7}{2}),$$

and P_1, P_3 are exponentially stable while P_2 is a saddle. The network being full-dimensional, all three lie in the unique compatibility class $\mathbb{R}^3_{\ge 0}$. Multistability and ACR persist for all rate constants in a neighbourhood of this vector.

Proof. The mass-action system is

$$\begin{aligned}\dot{x} &= 2k_4yz - k_5xz - 2k_6x^2,\\ \dot{y} &= k_1z - k_4yz + k_6x^2,\\ \dot{z} &= -k_2z^2 + k_3y - k_4yz + k_5xz.\end{aligned}$$

The reaction vectors $(0,1,0)$, $(0,0,-1)$, $(2,-1,-1)$ and $(-1,0,1)$ span $\mathbb{R}^3$. The weight $w = (1,2,0)$ gives $\dot{x} + 2\dot{y} = z\,(2k_1 - k_5x)$, so every positive steady state has $x = c := 2k_1/k_5$. On this plane $\dot{y} = 0$ gives $y = (k_1z + k_6c^2)/(k_4z)$, and then

$$k_4z\,\dot{z} = -k_2k_4z^3 + k_1k_4z^2 + (k_1k_3 - k_4k_6c^2)\,z + k_3k_6c^2,$$

which is positive at $z = 0$ and negative for large z. So a positive steady state exists for every κ, and ACR holds for every κ. For the given rate constants $c = 1$, $y = (5z+7)/(4z)$ and $4z\,\dot{z} = -(2z-1)(z-1)(2z-7)$, which gives exactly P_1, P_2, P_3. The Jacobian matrices and their characteristic polynomials are

$$J(P_1) = \begin{pmatrix} -33 & 4 & 28 \\ 14 & -2 & -14 \\ 5 & -1 & -10 \end{pmatrix}, \qquad \chi_1 = \lambda^3 + 45\lambda^2 + 206\lambda + 30 = (\lambda+5)(\lambda^2+40\lambda+6),$$
$$J(P_2) = \begin{pmatrix} -38 & 8 & 14 \\ 14 & -4 & -7 \\ 10 & -3 & -4 \end{pmatrix}, \qquad \chi_2 = \lambda^3 + 46\lambda^2 + 47\lambda - 50,$$
$$J(P_3) = \begin{pmatrix} -63 & 28 & 4 \\ 14 & -14 & -2 \\ 35 & -13 & -4 \end{pmatrix}, \qquad \chi_3 = \lambda^3 + 81\lambda^2 + 632\lambda + 1050.$$

By the Routh–Hurwitz criterion ($45 \cdot 206 > 30$ and $81 \cdot 632 > 1050$, all coefficients positive) P_1 and P_3 are exponentially stable; $\chi_2(0) = -50 < 0$ gives P_2 a positive real eigenvalue. All three steady states are nondegenerate, so they persist, with their stability types, for nearby rate constants, and ACR holds for all rate constants. □

The mechanism is a weighted integral feedback: the pair of complexes $\{Z,\ X + Z\}$ is the only one whose reactions change $w \cdot x = x + 2y$, so that $w \cdot f = z\,(2k_1 - k_5 x)$. Two further certified networks with six complexes, one of them weakly reversible with two linkage classes of deficiency zero and total deficiency one, are given in the sub-study `complexes/report.md`.

5 Five reactant complexes suffice

Consider four species A, B, C, R and the network $\mathcal{N}_2$ of six reactions

$$\begin{gathered} 0 \xrightarrow{64} C, \qquad A + B \xrightarrow{3} 2B, \qquad A + C \xrightarrow{5} B + C, \qquad B + C \xrightarrow{64} A, \\ 0 \xrightarrow{1} R, \qquad R \xrightarrow{1} 0. \end{gathered} \tag{22}$$

Its five distinct reactant complexes are $0, A + B, A + C, B + C, R$. Every reactant and product has molecularity at most two. With the mass-action convention kx^y its equations are

$$\begin{aligned} \dot a &= -3ab - 5ac + 64bc, & \dot b &= 3ab + 5ac - 64bc, \\ \dot c &= 64 - 64bc, & \dot r &= 1 - r. \end{aligned} \tag{23}$$

The stoichiometric subspace is $\{v : v_a + v_b = 0\}$, of dimension three; the only independent conservation law is $a + b = T$. Every positive equilibrium has $r = 1$, so ACR holds globally, across all compatibility classes. In fact arbitrary positive rates in the last two reactions give the ACR value k_{0R}/k_{R0}.

In the class $a + b = 9$, equilibrium requires $b = 9 - a$, $c = 1/(9 - a)$ and $r = 1$, with $0 < a < 9$. The remaining equation factors exactly as

$$\dot a\big|_{b=9-a,\,c=1/(9-a)} = \frac{-3(a-4)(a-6)(a-8)}{9-a} = 0.$$

Thus the complete list of positive equilibria in this class is

$$P_1 = (4, 5, 1/5, 1), \qquad P_2 = (6, 3, 1/3, 1), \qquad P_3 = (8, 1, 1, 1).$$

In the class coordinates (a, c, r) the restricted Jacobian is

$$J_{\mathrm{red}} = \begin{pmatrix} 6a - 27 - 69c & 576 - 69a & 0 \\ 64c & -64(9-a) & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Its characteristic polynomial is $(\lambda + 1)(\lambda^2 - \tau\lambda + \Delta)$, where τ and Δ are the trace and determinant of the upper left 2×2 block:

Equilibrium	τ	Δ	Stability in the class
P_1	$-1684/5$	1536	Exponentially stable
P_2	-206	-768	Saddle
P_3	-112	1536	Exponentially stable

The trace-negative, determinant-positive criterion for a real 2×2 matrix proves the two stability assertions exactly; no numerical eigenvalue calculation is used.

The regulator can also affect the other species without increasing the number of sources. Replace $0 \xrightarrow{64} C$, $0 \xrightarrow{1} R$, $R \xrightarrow{1} 0$ by $0 \xrightarrow{62} C$, $0 \xrightarrow{1} C + R$, $R \xrightarrow{1} C$. Only $\dot{c}$ changes, by adding $r - 1$. The equilibria are unchanged and the Jacobian remains triangular in the two blocks, with exactly the same characteristic polynomial. Both versions are verified directly from their reaction lists by `certify_five_example.py`.

6 Four reactant complexes do not suffice

Theorem 24. Let a mass-action system have stoichiometric rank three, four distinct reactant complexes, and at-most-bimolecular sources and products. If it has ACR, each positive compatibility class contains at most two nondegenerate positive equilibria. When there are two, their restricted Jacobian determinants have opposite signs. In particular, at most one is exponentially stable.

Corollary 25. An at-most-bimolecular mass-action system with ACR and two exponentially stable positive equilibria in one class has at least five distinct reactant complexes.

Proof. With r reactant complexes a positive equilibrium gives a strictly positive vector in the kernel of the aggregated coefficient matrix N, so $\operatorname{rank} N \leq r-1$. At a nondegenerate equilibrium, $\operatorname{im} N$ equals the stoichiometric subspace S: indeed $\operatorname{im} J \subseteq \operatorname{im} N \subseteq S$, while $J|_S$ is invertible. For $r \leq 4$ this forces $\dim S \leq 3$. Dimension two is excluded by Theorem 15, which holds for every number of species and reactant complexes. Dimension one is excluded by Proposition 3.10 of [1]; directly, on a class the field is a fixed vector times a polynomial of degree at most two in the class coordinate, and two simple zeros of such a polynomial have opposite derivative signs. Dimension zero has singleton classes. Dimension three forces $r = 4$ and is excluded by Theorem 24. □

Theorem 24 does not use Theorem 15; only Corollary 25 does.

7 Toric reduction and the determinant sign

List the sources as $y_1, \ldots, y_4$, and write $f(x) = Nm(x)$, where $m_a(x) = x^{y_a}$. It suffices to consider a system possessing a nondegenerate positive equilibrium x^*. Then $\operatorname{rank} N = 3$ and $\ker N$ is generated by $m(x^*) > 0$. Let A be the $3 \times n$ matrix with rows $y_a - y_4$, $a = 1, 2, 3$. The equilibrium equations are

$$A \log x = A \log x^*, \qquad \mathcal{E} = x^* \circ \exp(\ker A).$$

Nondegeneracy forces $\operatorname{rank} A = 3$. Otherwise the differential of these equations, and hence the restricted Jacobian, has rank at most two. ACR in species i is equivalent to $e_i \in \operatorname{row}(A)$, or to a zero ith row in any basis matrix V of $\ker A$.

Let W have rows forming a basis of $S^{\perp}$. Compatibility is $Wx = Wx^*$. Set $N_0 = [N_1\ N_2\ N_3]$ and $p_a = m_a(x^*)/m_4(x^*) > 0$. Any three columns of N are independent. At equilibrium,

$$J = m_4(x) N_0 \operatorname{diag}(p_1, p_2, p_3) A \operatorname{diag}(1/x).$$

Consequently, for a fixed nonzero constant σ and a positive factor $q(x)$,

$$\det(J|_S) = \sigma q(x) \det\begin{pmatrix} W \\ A\,\mathrm{diag}(1/x) \end{pmatrix}. \tag{26}$$

To see this, use the basis N_0 for S and complete it by columns B to a basis of $\mathbb{R}^n$. Multiplication of the displayed block matrix by $[B\ N_0]$ gives a block triangular matrix with diagonal blocks WB and $A\,\mathrm{diag}(1/x)N_0$. All change-of-basis determinants are fixed and nonzero.

The same determinant can be written with a basis matrix V of $\ker A$. Choose C with AC invertible. Then $[V\ C]$ is invertible, and multiplying the block matrix on the right by $\mathrm{diag}(x)[V\ C]$ gives a block triangular matrix with diagonal blocks $W\,\mathrm{diag}(x)V$ and AC. Hence, for a fixed constant $c \neq 0$,

$$\det\begin{pmatrix} W \\ A\,\mathrm{diag}(1/x) \end{pmatrix} = \frac{c}{x_1 \cdots x_n} \det(W\,\mathrm{diag}(x)V). \tag{27}$$

At a positive equilibrium, nondegeneracy is therefore equivalent to $\det(W\,\mathrm{diag}(x)V) \neq 0$, and the sign of $\det(J|_S)$ is a fixed sign times the sign of $\det(W\,\mathrm{diag}(x)V)$.

Replacing logarithmic binomials by ordinary binomials multiplies their differential rows by positive monomial factors, followed by a fixed nonsingular row transformation. Fixing the ACR concentration and performing linear eliminations therefore preserves the comparison of determinant signs. In a one-variable reduction $F(t) = 0$, the augmented determinant is a fixed sign times a positive factor times $F'(t)$, provided the eliminated equations have nonsingular differentials on the positive parameter interval. We verify this condition explicitly below. Two simple roots of a quadratic on an interval have opposite derivative signs. The same holds for $h(t) - p$ when $p > 0$ and $\log h$ is strictly concave on that interval. Since a real Hurwitz 3×3 matrix has negative determinant, opposite determinant signs preclude two stable equilibria.

8 Exhaustive source classification

A species absent from every source has a nonnegative derivative. At a positive equilibrium every reaction rate is positive, so no reaction can produce that species either. Such species are inert and can be removed. They cannot be ACR species, since their arbitrary positive concentrations give equilibria with different values. Four bimolecular sources then involve at most eight species.

The source classification is elementary and finite. For $4 \leq n \leq 8$, enumerate four-element subsets of

$$\{0, e_i, 2e_i, e_i + e_j : 1 \leq i < j \leq n\}.$$

Retain only sets covering all species, having rank $A = 3$, and having an ACR coordinate. The rank test uses integer 3×3 minors. The ACR coordinates are precisely the indices contained in every nonzero three-column minor: deleting such a column lowers the rank to two. Canonicalization permutes the four sources and sorts the species columns, so no floating-point rank decision or heuristic is involved. The script `certify_sources.py` performs this enumeration.

For three species the toric equilibrium set is a singleton. For four species there are 26 source types. In each, a generator v of $\ker A$ has at most three distinct entries, including zero; this is checked by its integer cofactor formula. The class equation is

$$\sum_{j=1}^{4} W_j x_j^* e^{v_j t} - W x^* = 0.$$

It has at most three distinct exponential terms. A nonzero sum of three real exponentials has at most two real zeros counted with multiplicity: divide by the smallest exponential and apply Rolle's theorem to the resulting derivative, which has at most two terms. If the sum is identically zero, all equilibria on the curve are degenerate. In the nondegenerate case the two determinants have opposite signs by (26).

For five or more species the exact list, up to species and source permutations, is as follows.

Type	Sources	ACR species
S1	$0, E, C+D, A+B$	E
S2	$0, 2E, C+D, A+B$	E
S3	$E, 2E, C+D, A+B$	E
S4	$E, D, C+E, A+B$	C
S5	$E, D+E, C+E, A+B$	C, D
S6	$E, D+E, C+D, A+B$	D
S7	$E, D+E, 2C, A+B$	D
S8	$E, D+E, B+C, A+C$	D
S9	$F, E+F, C+D, A+B$	E

There are eight types on five species, one on six species, and none on seven or eight.

For S4, S5, S6 and S8 the varying equilibrium concentrations, after positive constant factors are removed, belong to three groups $g_1 = u, g_2 = w/u, g_3 = w$. For S4, c is fixed and d, e are proportional to ab; for S5, c, d are fixed and e is proportional to ab; for S6, d is fixed and c, e are proportional to ab; for S8, d is fixed, a, b are proportional, and e is proportional to ac. The class equations are $Qg = T$, two linear equations in the three group variables. If $\operatorname{rank} Q < 2$, the equilibria are degenerate. Otherwise write the affine solution line as $g(t) = g^0 + t\rho$, where ρ spans $\ker Q$. Positivity of all three components cuts out an interval. The remaining equation is the polynomial $F(t) = g_1(t)g_2(t) - g_3(t)$ of degree at most two. The augmented determinant with rows Q and $\nabla(g_1 g_2 - g_3)$ is a fixed nonzero multiple of $F'(t)$: the cross product of the rows of Q is a fixed nonzero multiple of ρ. The group embedding is linear with independent columns and positive constant entries on disjoint coordinate groups. Eliminating the other fixed linear equations therefore changes the original augmented determinant only by a fixed nonzero factor. Hence (26) applies, and two nondegenerate equilibria have opposite determinant signs. If F vanishes identically, every equilibrium is degenerate.

9 Exhaustive stoichiometric classification

We now give an exact classification for S1, S2, S3, S7 and S9.

For a fixed source set, form every possible nonzero reaction vector $p - y_a$, where p is

any at-most-bimolecular product. Every rank-three stoichiometric subspace is spanned by three of these vectors. Enumerate these triples and identify equal spans by their primitive exterior coordinates (their 3×3 minors, divided by their gcd, with the first nonzero entry positive). For each span S, retain all allowed reaction vectors lying in it, giving a maximal matrix G. Membership is checked by the integer adjugate identity in a nonsingular three-coordinate projection.

Lemma 28. If some subnetwork with this span has a positive equilibrium, the maximal matrix G has a strictly positive kernel vector. In a three-coordinate projection, failure of this condition is detected by a nonzero cross product of two columns having nonnegative inner products with every column, after possibly reversing its sign.

Proof. The subnetwork equilibrium supplies a positive dependence among vectors spanning S. Their positive cone is therefore all of S: any linear representation can be made nonnegative by adding a sufficiently large multiple of that dependence. Each additional vector can thus be cancelled by a nonnegative combination of the old vectors. Adding these dependencies to the original positive dependence gives a dependence positive on every maximal-network reaction.

Conversely, a finite collection spanning $\mathbb{R}^3$ positively spans $\mathbb{R}^3$ exactly when its dual cone is $\{0\}$. If this dual cone is nonzero it is pointed, since the original vectors span $\mathbb{R}^3$. It has an extreme ray, on which at least two independent constraints are active. Such a ray is perpendicular to two independent original vectors, and hence is their cross product up to sign. This proves the exact finite test. □

Next write

$$D_W(z) = \det(W \operatorname{diag}(z) V) = \sum_{|I|=n-3} \det W_{:,I} \det V_{I,:} \prod_{i \in I} z_i. \tag{29}$$

If all coefficients vanish, every equilibrium is degenerate by (26) and (27). If the nonzero coefficients have one sign, there is at most one positive equilibrium per class. Indeed, if x, y are two such equilibria, $\log x - \log y = Vt$ and $x - y = \operatorname{diag}(\ell) V t$ with all logarithmic means $\ell_i > 0$. Then $W \operatorname{diag}(\ell) V t = 0$, contradicting (29) unless $t = 0$.

A further exact exclusion concerns the aggregated kinetic rank. Let K be a rational basis matrix of $\ker G$, write a general flux as $K\theta$, and sum its reaction vectors separately for each source, giving columns $L_1(\theta), \ldots, L_4(\theta)$ with sum zero. Project onto three independent coordinates. If the cubic polynomial

$$\det[L_1(\theta)\ L_2(\theta)\ L_3(\theta)]$$

vanishes identically, $\operatorname{rank} N < 3$ at every positive equilibrium of every subnetwork with this span. Such equilibria are degenerate. The certificate checks polynomial identity exactly. Nonzero evaluations are used only to retain a case, never to discard one.

The complete counts are:

	S1	S2	S3	S7	S9
All rank-three spans	4333	4287	4333	4397	23035
Missing a source	2965	3161	2965	3050	17734
No positive maximal flux	1126	782	1126	1042	4442
Injective	169	301	129	231	569
Class-degenerate	34	6	6	32	144
Kinetically degenerate	39	37	43	20	92
Remaining	0	0	64	22	54

Each span occurs in exactly one row after the first row. After fixing the ACR coordinate, remove its column from W and row-reduce. Permutations $A \leftrightarrow B$, $C \leftrightarrow D$, and interchange of the two pairs are allowed for S3 and S9; only $A \leftrightarrow B$ is needed for S7. The remaining spaces reduce to eleven, five and three forms respectively. The next sections list *all* of them and prove the required root and determinant properties.

10 The eleven S3 forms

The positive equilibria have

$$e = e_0, \qquad ab = p, \qquad cd = q, \qquad p, q, e_0 > 0.$$

After e is fixed the two conservation equations have one of the following forms. Constants T, U are arbitrary real numbers; existence of a positive solution supplies any needed positivity restrictions.

Index	First equation	Second equation
0	$b = T$	$c + d = U$
1	$a - 2c + 2d = T$	$b + c = U$
2	$a - c + d = T$	$b + (c + d)/2 = U$
3	$a - c + d = T$	$b + c = U$
4	$a - c + d = T$	$b + c + d = U$
5	$a - c + d = T$	$b + 2c = U$
6	$a - c/2 + d = T$	$b + c/2 = U$
7	$a = T$	$b + c + d = U$
8	$a + d = T$	$b + c = U$
9	$a + (c + d)/2 = T$	$b + (c + d)/2 = U$
10	$a + c + d = T$	$b + c + d = U$

Forms 0 and 7 fix $a, b > 0$ and $c + d$, so $cd = q$ is a quadratic on the interval $c, d > 0$. For form 8 take $a = t$, $b = p/t$, $c = U - p/t$, $d = T - t$. The positive interval is $p/U < t < T$, with $T, U > 0$. Multiplication of $cd - q = 0$ by $t > 0$ gives

$$-Ut^2 + (UT + p - q)t - pT = 0.$$

For forms 9 and 10, subtraction fixes $a - b$; together with $ab = p$ this uniquely fixes the positive pair (a, b), since $a - p/a$ is strictly increasing from $-\infty$ to $+\infty$. Again $c + d$ is fixed and $cd = q$ is quadratic on its positive interval.

For forms 1–6 put $c = t > 0$, $d = q/t$. Then

$$a = T + \alpha t - \beta q/t, \qquad b = U - \gamma t - \delta q/t,$$

where $\alpha, \beta, \gamma > 0$ and $\delta \geq 0$, as read directly from the table. The functions a, b are concave; a is strictly concave. Their common positivity set is an interval. On it,

$$(\log(ab))'' = \frac{a''}{a} - \frac{(a')^2}{a^2} + \frac{b''}{b} - \frac{(b')^2}{b^2} < 0.$$

Hence $ab = p$ has at most two roots, with opposite derivative signs when two roots exist.

For completeness, take the augmented determinant with rows R, $\nabla(ab - p)$, $\nabla(cd - q)$, where R is the conservation matrix in the table. On the displayed scalar curves its multiplier of the derivative is, up to a fixed sign: $T > 0$ for forms 0 and 7; $t > 0$ for forms 1–6 and 8; and $h(a + b) > 0$ for forms 9 and 10, where $h = 1/2$ and $h = 1$ respectively. For form 8 multiplication of the scalar equation by t preserves the sign at a root. These identities also prove that repeated roots and identically zero reductions cannot be nondegenerate equilibria.

11 The five S7 forms

Here the positive equilibrium equations are

$$d = d_0, \qquad ab = pe, \qquad c^2 = qe, \qquad p, q, d_0 > 0.$$

The reduced variables are (a, b, c, e), and the conservation equations are:

Index	First equation	Second equation
0	$b + e = T$	$c - e = U$
1	$a - c + 2e = T$	$b + c = U$
2	$a - c/2 + e/2 = T$	$b + c/2 + e/2 = U$
3	$a - e = T$	$b = U$
4	$a + b + e = T$	$c = U$

For form 0 take $c = t$, $e = t - U$, $b = T - t + U$, $a = pe/b$. Positivity is an interval, and $c^2 - qe = t^2 - q(t - U) = 0$ is quadratic. Its determinant multiplier has magnitude $b > 0$.

For forms 1 and 2 put $u = 1/c > 0$, so $e = 1/(qu^2)$. Then

$$A(u) := a/c = Tu + \alpha - \frac{\beta}{qu}, \qquad B(u) := b/c = Uu - \gamma - \frac{\delta}{qu},$$

with $(\alpha, \beta, \gamma, \delta) = (1, 2, 1, 0)$ or $(1/2, 1/2, 1/2, 1/2)$. The positive interval is $A(u) > 0$, $B(u) > 0$. Both functions are concave and $A''(u) < 0$. Thus $\log(AB)$ is strictly concave, and $AB = p/q$ has at most two roots with opposite derivative signs. If $H = AB - p/q$, the augmented determinant with rows R, $\nabla(ab - pe)$, $\nabla(c^2 - qe)$ equals, up to fixed orientation,

$$q(H'(u) - 2H(u)/u).$$

At a root its multiplier of H' is the positive constant q.

For form 3 the equation $ab = pe$ is $UT + (U - p)e = 0$, with $U = b > 0$. There is at most one positive solution, unless the equation is an identity, in which case the positive solutions form a curve and are degenerate. For form 4, $c = U > 0$ fixes $e = U^2/q$; the sum and product of a, b are fixed, giving a quadratic on $a, b > 0$. The determinant multiplier has magnitude $q > 0$ in both scalar parametrizations used by the certificate.

12 The three S9 forms

The equilibrium equations are

$$e = e_0, \qquad ab = pf, \qquad cd = qf, \qquad p, q, e_0 > 0.$$

In variables (a, b, c, d, f) the three conservation equations have one of these forms:

Index	First equation	Second equation	Third equation
0	$b + f = T$	$c - f = U$	$d = V$
1	$a - c/2 + f/2 = T$	$b + c/2 + f/2 = U$	$d = V$
2	$a = T$	$b = U$	$c + d + f = V$

For form 0, put $f = t$, $b = T - t$, $c = t + U$, $d = V$, $a = pt/(T - t)$. Positivity is an interval with $V > 0$. The remaining equation is $(V - q)t + VU = 0$. It has at most one solution, or is an identity giving only degenerate equilibria. The determinant multiplier has magnitude $b > 0$.

For form 1, put $f = t$, $d = V > 0$, $c = qt/V$,

$$a = T + \tfrac{1}{2}\,(q/V - 1)t, \qquad b = U - \tfrac{1}{2}\,(q/V + 1)t.$$

The positive domain is an interval defined by three linear inequalities $t, a, b > 0$. The remaining equation $ab - pt = 0$ is quadratic. Its determinant multiplier has magnitude $V > 0$.

For form 2, $a = T > 0$ and $b = U > 0$ fix $f = TU/p$. The sum $c + d = V - TU/p$ and product $cd = qTU/p$ are fixed, so there are at most two positive solutions, with opposite derivative signs. The determinant multiplier has magnitude $p > 0$.

All nineteen forms have now been treated. Together with the source classification, the injectivity and degeneracy exclusions, and (26), this proves Theorem 24.

13 Further results and open questions

The following results complement Table 2. Their proofs, partly computer-assisted with exact arithmetic, are in the two sub-study reports `examples/report.md` and `complexes/report.md`, with the scripts that check them.

(1) *A second three-species example with rational steady states.* The network $\mathcal{G}_7 = \{X + Y \xrightarrow{1} 2Y,\ Y \xrightarrow{1} 2Z,\ 0 \xrightarrow{5} X + Z,\ X + Z \xrightarrow{4} 2Z,\ 2Z \xrightarrow{1} X + Z,\ Y + Z \xrightarrow{6} Y\}$ has $\dot{y} = y(k_1 x - k_2)$, hence ACR in X for all rate constants, and for the indicated ones exactly three positive steady states $(1, 2, 1)$, $(1, \frac{5}{4}, \frac{3}{2})$, $(1, \frac{10}{9}, \frac{5}{3})$, the first and the last exponentially stable (exact Gröbner basis and Routh–Hurwitz). It has 7 complexes.

(2) *Six complexes in two linkage classes.* The weakly reversible network $\{2Z \rightleftarrows X + Z,\ Y + Z \to X,\ Y + Z \to X + Y,\ X \to 2X,\ X + Y \to X,\ 2X \to Y + Z\}$ has 6 complexes, two linkage classes of deficiency 0 and total deficiency 1, ACR in X and, for suitable rate constants, two exponentially stable positive steady states. So the deficiency-one theorem cannot be applied to two linkage classes when $\sum_\theta \delta_\theta < \delta$, and six-complex examples need not have a single linkage class.

(3) *Reactions with three species.* No full-dimensional bimolecular network with 3 species and at most 5 reactions has three nondegenerate positive steady states, or two exponentially stable ones, for any rate constants. Networks with at most 4 distinct reactant complexes are covered by Propositions 3.1(3) and 3.5 [1]; the 3 188 646 networks with 5 distinct reactant complexes were enumerated exactly, through a reduction to the sign changes of one univariate function whose derivative has, up to a fixed sign, the sign of the Jacobian determinant (two exponentially stable states have determinants of the same sign, which forces a third steady state between them). By Corollary 17, a three-species network with ACR and multistability is full-dimensional, so it needs at least 6 reactions, which $\mathcal{N}_1$ and $\mathcal{G}_7$ attain. The restriction to full-dimensional networks is necessary: the two-dimensional network $\{0 \xrightarrow{10} Z,\ Y+Z \xrightarrow{10} X,\ X+Z \xrightarrow{1} Y+Z,\ X+Y \xrightarrow{1} 2Y\}$ has the three nondegenerate positive steady states $(5,1,1)$, $(4,2,\frac{1}{2})$, $(3,3,\frac{1}{3})$ in the class $x+y=6$, the outer two exponentially stable (it has no ACR).

Open questions. Does one network attain all four minima of Table 2? $\mathcal{N}_1$ has six reactant complexes and $\mathcal{N}_2$ has four species. In particular, for full-dimensional networks on three species, is the bound $n+2=5$ of Theorem 5.1 [1] attained with multistability? Does Theorem 15 hold for networks with bimolecular reactant complexes but larger products (Remark 16)? Finally, [5] found ACR with bistability in a dual-site futile cycle with bifunctional enzymes (six species, dimension four); our examples show that much smaller mechanisms, based on a weighted integral feedback, suffice.

14 Computer verification

All scripts print `PASS`/`FAIL` for each claim and exit with a nonzero code on failure. They were run under WSL (Fedora 44) with Python 3.12, sympy 1.14, python-flint 0.9 and NumPy.

- `scripts/check_dimension_two_example.py` (8 s): on 20 random rational instances of the KNS network of Example 4.10, checks the determinant formula of Lemma 10 and that the two positive steady states of a class have determinants of opposite signs. This is a sanity check; Theorem 15 is proved by hand.
- `complexes/scripts/certify_examples.py` (exact rational arithmetic, standard library only, under 0.1 s): Theorem 21 and the two further six-complex examples (ACR identity, steady states, characteristic polynomials, Routh–Hurwitz conditions, counts of complexes and deficiency). `complexes/scripts/eigen_enclosures.py`: eigenvalue enclosures in ball arithmetic.
- `examples/scripts/certify_example.py` (under 1 s): the network $\mathcal{G}_7$ (exact Gröbner basis, root isolation, ball-arithmetic eigenvalues). The enumeration of item 3 is `examples/scripts/enum5_reactions.py` (about two minutes on 12 cores).

Searches and simulations used to find the examples are in `explore` sub-folders; they are floating-point and not part of any proof.

For the four-reactant classification and the network $\mathcal{N}_2$, run from the folder `four-reactants/`:

```
python scripts/certify_sources.py
```

```
python scripts/certify_subspaces.py
python scripts/certify_normal_forms.py
python scripts/certify_five_example.py
```

The first script re-enumerates every eligible source set. The second generates from scratch all 40,385 reaction spans for the five specified source types, proves each exclusion, and writes the retained spaces and their classifications to `certificates/subspaces.json`. It does not read any numerical candidate list. The third symbolically checks all nineteen determinant identities and the concavity derivatives, including the conservation equations along every scalar curve. The fourth derives the upper-bound example from its reaction list, factors its complete steady-state equation and checks every stability claim in rational arithmetic. Each script prints PASS/FAIL and exits nonzero on a failure.

The environment used was Python 3.12, SymPy 1.14 and NumPy under Fedora 44 in WSL. The runs take less than one minute in total. Integer arrays use signed 64-bit entries; their input values are in $[-2, 2]$, three-minors have absolute value at most 48, adjugate entries at most 8, membership products at most 288, and cross-product inner products at most 48. Thus these operations cannot overflow. Nullspaces, polynomial identities and all subsequent coefficients are rational and symbolic. No numerical eigenvalue, optimizer tolerance, random search or sampling result is used as an exclusion.

Fresh-context verifiers checked the two reports merged here; see `verification.md` and `four-reactants/verification.md`. The second one reimplemented the enumeration, obtained the same label for every span, and recorded two facts that make the proof sturdier. The positive-flux exclusion is not needed: every span it rejects is also rejected by the injectivity or degeneracy tests. The kinetic exclusion also has an elementary reason, without the cubic: on each span that it removes, the spans of the reaction vectors of some three sources fail Rado's condition, so three columns of N are forced to be dependent.

The merged text itself has not been re-verified. A public-source novelty audit on 31 October 2026, recorded in `four-reactants/references.md`, found no prior resolution of the exact-minimum question; absence of search hits is not a proof of priority. This is a computer-assisted proof, checked by fresh-context AI verifiers, not by human referees. The results do not concern larger products or nonhyperbolic notions of stability, and they do not classify every network attaining the minima.

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